UNIT 5 · ARS/NET PATTERN

MCQ Bank — 140 Questions with Answers

Anthropometry, physiological cost of work, fatigue and comfort (Q1–Q35)

Q1. Anthropometry is the science of:

(a) Muscle chemistry
(b) Human body dimensions and strengths
(c) Soil physics
(d) Plant growth

Q2. Standard workplace-design practice uses which percentiles?

(a) 25th & 75th
(b) 5th & 95th
(c) 50th only
(d) 1st & 99th

Q3. Clearance dimensions (cab height, doorways) are designed for the:

(a) 5th percentile
(b) 50th
(c) 95th percentile
(d) Mean

Q4. Reach dimensions (controls, pedals) are designed for the:

(a) 5th percentile
(b) 95th
(c) 99th
(d) Tallest user

Q5. Per Liljedahl (p. 227), inside cab height to protect 95% of male drivers is:

(a) 160 cm
(b) 170 cm
(c) 185 cm
(d) 200 cm

Q6. The recommended anthropometric practice for agricultural equipment in the US is:

(a) SAE J833
(b) IS 4468
(c) ISO 500
(d) OECD

Q7. Seat height in workplace design is based on:

(a) Stature
(b) Popliteal height
(c) Elbow breadth
(d) Mass

Q8. Seat pan depth is based on:

(a) Buttock–popliteal length
(b) Knee height
(c) Sitting height
(d) Stature

Q9. The 50th-percentile mass of the SAE male population (Table 9-5) is about:

(a) 55 kg
(b) 65 kg
(c) 75 kg
(d) 90 kg

Q10. The best design strategy for spanning users' sizes is:

(a) Design for the average
(b) Adjustability from 5th to 95th percentile
(c) One size
(d) Design for tallest

Q11. Indian equipment design uses CIAE/AICRP anthropometric data because:

(a) US data are secret
(b) Indian body dimensions are smaller — western data oversize designs
(c) Fashion
(d) Cost

Q12. For continuous work, sustained force should not exceed about what share of maximum voluntary strength?

(a) 90% (b) 60%
(c) 30% (d) 5%

Q13. One litre of oxygen consumed corresponds to about:

(a) 5 kJ (b) 10 kJ
(c) 20.9 kJ (d) 50 kJ

Q14. The practical field index of physiological cost is:

(a) Blood sugar (b) Heart rate (telemetry)
(c) EEG (d) Height

Q15. Oxygen uptake in field studies is measured with:

(a) Anemometer (b) Douglas bag / portable respirometer
(c) Hydrometer (d) Dynamometer

Q16. Using E(kJ/min) = 0.159·HR − 8.72, HR = 130 gives about:

(a) 6 (b) 9
(c) 11.9 (d) 20 kJ/min

Q17. Acceptable workload (AWL) for an 8-h day is about:

(a) 15% (b) 35% of VO_2max
(c) 70% of VO_2max (d) 100%

Q18. The limit of continuous performance corresponds to a working heart rate of about:

(a) 90 (b) 110
(c) 140 (d) 170 beats/min

Q19. The recommended increase of working HR over resting (ΔHR) is at most about:

(a) 10 (b) 25
(c) 40 (d) 80 beats/min

Q20. VO_2max of an average Indian male agricultural worker is about:

(a) 1.0 (b) 2.0
(c) 3.5 (d) 5.0 L/min

Q21. On the Christensen scale, 'moderate' work is:

(a) <2.5 (b) 2.5–5
(c) 5–7.5 (d) 10–12.5 kcal/min

Q22. Work above ~12.5 kcal/min is graded:

(a) Light (b) Moderate
(c) Heavy (d) Extremely heavy

Q23. STATIC muscular work fatigues faster than dynamic because:

(a) It is boring (b) Sustained contraction occludes blood flow to the muscle
(c) More sweat (d) Less skill

Q24. Holding a sprayer lance for long periods is an example of:

(a) Dynamic work (b) Static work
(c) Aerobic exercise (d) Rest

Q25. Murrell's rest allowance is R =

(a) T(b−s)/(b−1.5) (b) T·b/s
(c) (b+s)/T (d) T−b−s

Q26. In Murrell's formula, 1.5 kcal/min represents:

(a) Standard work rate
(b) Resting metabolic rate
(c) Maximum rate
(d) Sleep rate

Q27. For b = 8 and s = 4 kcal/min, rest fraction ≈

(a) 20%
(b) 40%
(c) 62%
(d) 80%

Q28. Better fatigue management favours:

(a) One long break
(b) Frequent short pauses and task rotation
(c) No breaks
(d) Faster pace

Q29. The 10-point subjective scale for whole-body discomfort is the:

(a) BIS scale
(b) ODR (overall discomfort rating)
(c) pH scale
(d) Beaufort scale

Q30. Localised discomfort is mapped using the:

(a) Body part discomfort score (Corlett–Bishop)
(b) Nebraska chart
(c) MWD
(d) NDVI

Q31. ASHRAE effective-temperature comfort zone (Liljedahl p. 206) is about:

(a) 15–18 °C
(b) 24–27 °C
(c) 30–35 °C
(d) 37–40 °C

Q32. The dust-concentration guideline for combine cabs cited (p. 204) is:

(a) 1 mg/m³
(b) 15 mg/m³
(c) 150 mg/m³
(d) 500 mg/m³

Q33. A tractor cab pressurized to 50 Pa with fine filters reduced mean breathing-zone dust from 146 to about:

(a) 100
(b) 75
(c) 24.7
(d) 2 mg/m³

Q34. Effect of rising fatigue on work is:

(a) More output
(b) Falling output & accuracy, more errors and accidents
(c) No change
(d) Better vigilance

Q35. Physiological responses to work include:

(a) Heart rate and oxygen uptake
(b) Pulmonary ventilation and body temperature
(c) Muscle EMG activity
(d) All of these

Answers

Q1 — (b) Human body dimensions and strengths
Q2 — (b) 5th & 95th
Q3 — (c) 95th percentile — *Liljedahl pp. 227–229 — clearance from the upper percentile.*
Q4 — (a) 5th percentile — *Liljedahl p. 229 — reach from the lower percentile.*
Q5 — (c) 185 cm — *Liljedahl p. 227 — the 185-cm cab-height example.*
Q6 — (a) SAE J833
Q7 — (b) Popliteal height — *Table 9-5, p. 228 (SAE J833 data).*
Q8 — (a) Buttock–popliteal length
Q9 — (c) 75 kg — *Table 9-5, p. 228 — 50th-percentile mass 75 kg.*
Q10 — (b) Adjustability from 5th to 95th percentile
Q11 — (b) Indian body dimensions are smaller — western data oversize designs
Q12 — (c) 30%

Q13 — (c) 20.9 kJ
Q14 — (b) Heart rate (telemetry)
Q15 — (b) Douglas bag / portable respirometer
Q16 — (c) 11.9 — *0.159 × 130 − 8.72 = 11.95 kJ/min ≈ 2.85 kcal/min.*
Q17 — (b) 35% of VO_2max
Q18 — (b) 110
Q19 — (c) 40
Q20 — (b) 2.0
Q21 — (c) 5–7.5
Q22 — (d) Extremely heavy
Q23 — (b) Sustained contraction occludes blood flow to the muscle
Q24 — (b) Static work
Q25 — (a) T(b−s)/(b−1.5)
Q26 — (b) Resting metabolic rate
Q27 — (c) 62% — *R/T = (8 − 4)/(8 − 1.5) = 4/6.5 = 61.5%.*
Q28 — (b) Frequent short pauses and task rotation
Q29 — (b) ODR (overall discomfort rating)
Q30 — (a) Body part discomfort score (Corlett–Bishop)
Q31 — (b) 24–27 °C — *pp. 204–206 — Table 9-1 comfort zones; ET* 24–27 °C.*
Q32 — (b) 15 mg/m³ — *p. 204 — 15 mg/m³ guideline (Netherlands).*
Q33 — (c) 24.7 — *p. 204 — 577/146 → 24.7 mg/m³ with a 50-Pa pressurized cab.*
Q34 — (b) Falling output & accuracy, more errors and accidents
Q35 — (d) All of these

Ergonomics of farm tools; noise and vibration (Q36–Q70)

Q36. The audible frequency range of the human ear is:

(a) 1–100 Hz
(b) 16–20,000 Hz
(c) 100–1000 Hz
(d) 20,000–50,000 Hz

Q37. The reference sound pressure for airborne sound (0 dB) is:

(a) 0.02 N/m²
(b) 0.00002 N/m²
(c) 2 N/m²
(d) 0.2 N/m²

Q38. Sound pressure level is defined as:

(a) $10 \cdot \log(p/p_0)$
(b) $20 \cdot \log(p/p_0)$
(c) p/p_0
(d) $p - p_0$

Q39. The 'A' in dB(A) denotes:

(a) Amplifier
(b) Frequency weighting mimicking the human ear
(c) Average
(d) Absolute

Q40. Two equal 90-dB sources together give:

(a) 90 dB
(b) 93 dB
(c) 96 dB
(d) 180 dB

Q41. Per OSHA (Liljedahl Table 9-2), 90 dB(A) exposure is allowed for:

(a) 2 h
(b) 4 h
(c) 8 h
(d) Unlimited

Q42. Each 5-dB(A) increase in level:

(a) Doubles allowed time
(b) Halves the permissible exposure time
(c) No change
(d) Quadruples it

Q43. Permissible daily exposure at 105 dB(A) is:

(a) 4 h
(b) 2 h
(c) 1 h
(d) 15 min

Q44. Impulse/impact noise must not exceed a peak of:

(a) 90 dB
(b) 115 dB
(c) 140 dB
(d) 200 dB

Q45. By 1984 Nebraska tests, tractors WITH cabs averaged operator noise of about:

(a) 95 dB(A)
(b) 90 dB(A)
(c) 79.6 dB(A)
(d) 70 dB(A)

Q46. Manufacturers reduced operator noise primarily by:

(a) Smaller engines
(b) Sound-engineered cabs
(c) Slower speeds
(d) Bigger mufflers only

Q47. Chronic noise exposure above limits causes:

(a) Temporary deafness only
(b) Permanent noise-induced hearing loss
(c) Better hearing
(d) No effect

Q48. An immediate work-performance effect of noise is:

(a) Better focus
(b) Masking of speech and warning signals
(c) Cooling
(d) Higher torque

Q49. The first priority in noise control is:

(a) Ear plugs
(b) Reduction at the source
(c) Warning signs
(d) Shorter breaks

Q50. Ride vibration intensity generally increases with:

(a) Lower tyre pressure
(b) Ground speed
(c) Engine oil grade
(d) Paint

Q51. Matthews (1972) found most large tractors operated below 2/3 maximum power because:

(a) Weak engines
(b) Operators could not withstand the ride at full speed
(c) Fuel cost
(d) Law

Q52. Rossegger's X-ray study found spine deformations in what share of tractor drivers?

(a) 25%
(b) 43%
(c) 71.3%
(d) 95%

Q53. The only occupation worse than tractor driving for spines in that study:

(a) Miners
(b) Truck drivers
(c) Bus drivers
(d) Builders

Q54. The international guide for human whole-body vibration exposure is:

(a) ISO 500
(b) ISO 2631
(c) ISO 730
(d) ISO 9001

Q55. The seated human is most sensitive to vertical vibration at:

(a) 0.1–1 Hz
(b) 4–8 Hz
(c) 50–100 Hz
(d) 1000 Hz

Q56. Undamped bounce natural frequency of wheel tractors is about:

(a) 0.5–1 Hz (b) 3–5 Hz
(c) 15–20 Hz (d) 50 Hz

Q57. ISO 2631 'FDP' boundaries stand for:

(a) Full design power (b) Fatigue-decreased proficiency
(c) Field data points (d) Free damping period

Q58. Seat transmissibility is defined as:

(a) Input/output vibration (b) Output/input vibration intensity
(c) Spring rate (d) Damping ratio

Q59. Satisfactory seat transmissibility is about:

(a) 1.5–2 (b) 1.0
(c) 0.4–0.65 (d) 0.05

Q60. The seat's natural frequency should be what fraction of the tyres' natural frequency?

(a) 2× (b) Equal
(c) 0.4–0.5 (d) 0.1

Q61. Vibration isolation occurs only when the frequency ratio exceeds:

(a) 0.5 (b) 1
(c) √2 (d) 3

Q62. Seat suspensions include which elements?

(a) Variable-rate springs (b) Shock absorbers
(c) Weight adjustment for the operator (d) All of these

Q63. Hand-arm vibration exposure is assessed per:

(a) ISO 5349 (b) ISO 11783
(c) IS 9020 (d) SAE J833

Q64. 'Vibration white finger' results from:

(a) Whole-body vibration (b) Prolonged hand-transmitted vibration
(c) Cold water (d) Noise

Q65. A well-known Indian HAV source is:

(a) Seed drill (b) Power tiller handlebars
(c) MB plough (d) Level

Q66. Recommended diameter of a power-grip cylindrical tool handle is:

(a) 10–15 mm (b) 30–40 mm
(c) 60–80 mm (d) 100 mm

Q67. Long-handled weeders (wheel hoe) mainly reduce:

(a) Cost (b) Bending posture and spinal/energy load
(c) Weight (d) Blade wear

Q68. Ergonomic evaluation of an improved tool measures:

(a) Output capacity (b) Heart rate/energy cost
(c) ODR/BPDS discomfort (d) All of these

Q69. CIAE improved hand tools typically achieve:

(a) Lower output (b) Higher output at lower physiological cost
(c) Same everything (d) Only lower cost

Q70. Subtracting a small source (81 dB) from a 90-dB total leaves about (GenRad chart, p. 211):

(a) 9 dB (b) 81 dB

(c) 89.4 dB (d) 45 dB

Answers

Q36 — (b) 16–20,000 Hz — *Liljedahl p. 207 — ear range.*

Q37 — (b) 0.00002 N/m² — *Liljedahl p. 209 — p_0 = 0.00002 N/m².*

Q38 — (b) 20·log(p/p_0) — *Liljedahl p. 209 — SPL = 20 log p/p_0.*

Q39 — (b) Frequency weighting mimicking the human ear

Q40 — (b) 93 dB

Q41 — (c) 8 h — *OSHA Table 9-2, p. 208 — the 90-dB(A)/8-h line.*

Q42 — (b) Halves the permissible exposure time — *Table 9-2 — the 5-dB halving rule.*

Q43 — (c) 1 h — *90 → 8 h, 95 → 4 h, 100 → 2 h, 105 → 1 h.*

Q44 — (c) 140 dB — *Impulse noise limit 140 dB peak.*

Q45 — (c) 79.6 dB(A) — *p. 207 — 1984 data: no cab 94.0 vs cab 79.6 dB(A).*

Q46 — (b) Sound-engineered cabs — *Quiet came from cab engineering.*

Q47 — (b) Permanent noise-induced hearing loss

Q48 — (b) Masking of speech and warning signals

Q49 — (b) Reduction at the source

Q50 — (b) Ground speed

Q51 — (b) Operators could not withstand the ride at full speed — *pp. 210 — Matthews' two-thirds-power finding.*

Q52 — (c) 71.3% — *Rossegger, Table 9-3, p. 212 — 71.3%.*

Q53 — (b) Truck drivers — *Truck drivers 80%.*

Q54 — (b) ISO 2631

Q55 — (b) 4–8 Hz

Q56 — (b) 3–5 Hz — *p. 213 — tractor bounce 3–5 Hz.*

Q57 — (b) Fatigue-decreased proficiency

Q58 — (b) Output/input vibration intensity

Q59 — (c) 0.4–0.65 — *pp. 219, 223 — transmissibility target.*

Q60 — (c) 0.4–0.5 — *p. 223 — seat-to-tyre frequency ratio.*

Q61 — (c) √2

Q62 — (d) All of these

Q63 — (a) ISO 5349

Q64 — (b) Prolonged hand-transmitted vibration

Q65 — (b) Power tiller handlebars

Q66 — (b) 30–40 mm

Q67 — (b) Bending posture and spinal/energy load

Q68 — (d) All of these

Q69 — (b) Higher output at lower physiological cost

Q70 — (c) 89.4 dB — *p. 211 — difference 9 dB → correction 0.6 → 90 - 0.6 = 89.4 dB.*

Machinery safety, chemical hazards and protective gadgets (Q71–Q105)

Q71. Per Liljedahl, the MOST positive method of improving safety is:

(a) Education (b) Regulation

(c) Designing the hazard out of the product (d) Insurance

Q72. A safety shield that is inconvenient will most likely be:

(a) Praised
(b) Discarded by the operator
(c) Mandatory
(d) Cheaper

Q73. In US 1982 data, overturns accounted for what share of on-farm tractor fatalities?

(a) 20%
(b) 33%
(c) 49%
(d) 75%

Q74. Run-over deaths accounted for about:

(a) 5%
(b) 20%
(c) 40%
(d) 60% of tractor fatalities

Q75. In more than half of run-over cases the victim had first:

(a) Fallen from the tractor
(b) Been asleep
(c) Been on foot far away
(d) Been in a cab

Q76. Deaths per 100,000 tractors fell from 17.6 (1969) to what value in 1982?

(a) 15.2
(b) 12.0
(c) 8.1
(d) 2.0

Q77. That fall is attributed mainly to:

(a) Slower tractors
(b) Increasing ROPS fitment and seat-belt use
(c) Fewer tractors
(d) Better roads

Q78. Which is NOT one of the three basic ROPS types?

(a) Skeleton frame
(b) Frame with overhead roof
(c) Enclosed cab
(d) Roll cage of fabric

Q79. ROPS for wheeled agricultural tractors is covered by:

(a) ASAE S383.1
(b) ASAE S209
(c) ISO 500
(d) IS 6284

Q80. Overhead protection test procedures are covered by:

(a) S310.3
(b) J833
(c) IS 4931
(d) ISO 730

Q81. In ROPS testing, the static or dynamic loading test is followed by a:

(a) Paint test
(b) Static crush test on the same ROPS
(c) Fuel test
(d) Noise test

Q82. The field upset test may be omitted if:

(a) The tractor is small
(b) The laboratory test indicates compliance
(c) It rains
(d) Never

Q83. ROPS protects the operator ONLY if:

(a) The cab is closed
(b) The seat belt is fastened
(c) Lights are on
(d) Speed is high

Q84. Besides overturn, ROPS with roof protects against:

(a) Falling objects
(b) Noise only
(c) Fuel loss
(d) Dust only

Q85. The largest single cause of machinery injuries in Indian farms is associated with:

(a) Seed drills
(b) Threshers (cylinder entanglement while feeding)
(c) Ploughs
(d) Levellers

Q86. The Indian standard prescribing safe thresher feeding-chute geometry is:

(a) IS 9129 (b) IS 6316

(c) IS 4468 (d) IS 2062

Q87. Safety requirements for power threshers are in:

(a) IS 9020 (b) ISO 500

(c) IS 1239 (d) SAE J709

Q88. PTO shaft safety requires:

(a) Only paint (b) Master shield plus freely-rotating tunnel guard

(c) A chain (d) Nothing

Q89. Loose clothing near a rotating PTO shaft chiefly risks:

(a) Cooling (b) Entanglement and severe limb injury

(c) Corrosion (d) Noise

Q90. A hydraulic pinhole leak should NEVER be located with:

(a) Cardboard (b) Bare hands (injection injury risk)

(c) Goggles on (d) Engine off

Q91. Before working under a raised implement one must:

(a) Trust the hydraulics (b) Block/support it mechanically

(c) Increase pressure (d) Idle the engine

Q92. A neutral-start interlock ensures the engine starts only when:

(a) PTO is engaged (b) Transmission is in neutral/PTO off

(c) Lights are on (d) Fuel is full

Q93. LD_{50} means the dose that:

(a) Cures 50% (b) Kills 50% of test animals

(c) Is 50 mg (d) Lasts 50 days

Q94. A LOWER LD_{50} value indicates the chemical is:

(a) Safer (b) More toxic

(c) Cheaper (d) Less volatile

Q95. The RED triangle label denotes oral LD_{50} of:

(a) 1–50 mg/kg (extremely toxic) (b) 51–500

(c) 501–5000 (d) >5000

Q96. The YELLOW triangle corresponds to:

(a) Extremely toxic (b) Highly toxic, LD_{50} 51–500

(c) Moderately toxic (d) Slightly toxic

Q97. The BLUE triangle signal word is:

(a) POISON (b) DANGER

(c) CAUTION (d) SAFE

Q98. The GREEN triangle indicates oral LD_{50}:

(a) <50 (b) 51–500

(c) 501–5000 (d) >5000 mg/kg

Q99. Pesticide labelling classes in India are prescribed under the:

(a) Insecticides Act/Rules (b) Factories Act

(c) Motor Vehicles Act (d) BIS Act

Q100. The MAJOR route of pesticide entry during spraying is:

(a) Oral
(b) Dermal (skin)
(c) Ocular
(d) Rectal

Q101. Antidote for organophosphate poisoning is:

(a) Atropine
(b) Glucose
(c) Vitamin C
(d) Antihistamine

Q102. Blocked spray nozzles should be cleared:

(a) By blowing with the mouth
(b) With a pin/brush or water
(c) With petrol
(d) With the tongue

Q103. Which gloves are correct for pesticide handling?

(a) Cotton
(b) Nitrile/rubber
(c) Woollen
(d) None

Q104. The waiting (pre-harvest) interval is observed to:

(a) Save chemical
(b) Ensure residues fall below safe limits before harvest
(c) Improve yield
(d) Reduce cost

Q105. In the hazard-control hierarchy, PPE ranks:

(a) First
(b) Second
(c) Third
(d) Last — after elimination, engineering and administrative controls

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Answers

Q71 — (c) Designing the hazard out of the product — *Liljedahl p. 234 — design that removes the hazard is the most positive method.*

Q72 — (b) Discarded by the operator — *pp. 234–235 — an inconvenient shield is soon discarded.*

Q73 — (c) 49% — *p. 231 (NSC Accident Facts 1983, for 1982).*

Q74 — (b) 20%

Q75 — (a) Fallen from the tractor — *More than half had fallen from the tractor first.*

Q76 — (c) 8.1 — *Rate 17.6 → 8.1 per 100,000 tractors.*

Q77 — (b) Increasing ROPS fitment and seat-belt use

Q78 — (d) Roll cage of fabric — *p. 232 — the three ROPS types.*

Q79 — (a) ASAE S383.1

Q80 — (a) S310.3

Q81 — (b) Static crush test on the same ROPS — *p. 232 — static or dynamic → crush → field upset (omittable) → temperature–material.*

Q82 — (b) The laboratory test indicates compliance

Q83 — (b) The seat belt is fastened

Q84 — (a) Falling objects

Q85 — (b) Threshers (cylinder entanglement while feeding)

Q86 — (a) IS 9129

Q87 — (a) IS 9020

Q88 — (b) Master shield plus freely-rotating tunnel guard

Q89 — (b) Entanglement and severe limb injury

Q90 — (b) Bare hands (injection injury risk)

Q91 — (b) Block/support it mechanically

Q92 — (b) Transmission is in neutral/PTO off

Q93 — (b) Kills 50% of test animals

Q94 — (b) More toxic — *LD_{50} is an inverse measure of toxicity.*

Q95 — (a) 1–50 mg/kg (extremely toxic) — *The four label triangles are a guaranteed-recall item.*

Q96 — (b) Highly toxic, LD_{50} 51–500

Q97 — (b) DANGER

Q98 — (d) >5000 mg/kg

Q99 — (a) Insecticides Act/Rules

Q100 — (b) Dermal (skin) — *Skin absorption dominates in knapsack spraying, so gloves/overalls matter more than masks in many cases.*

Q101 — (a) Atropine

Q102 — (b) With a pin/brush or water

Q103 — (b) Nitrile/rubber — *Cotton absorbs and holds the chemical against the skin.*

Q104 — (b) Ensure residues fall below safe limits before harvest

Q105 — (d) Last — after elimination, engineering and administrative controls

Tractor controls, visual range and seat design (Q106–Q140)

Q106. The recommended seat-cushion (pan) angle from horizontal is:

(a) 0° (b) 5° with front raised

(c) 15° (d) 25° rear raised

Q107. The angle between backrest and seat should be approximately:

(a) 75° (b) 90° exactly

(c) 95° (minimum 90°) (d) 120°

Q108. The backrest must specifically support the:

(a) Neck only (b) Lumbar back

(c) Shoulders only (d) Head

Q109. For pedal operation the leg-to-thigh angle should be:

(a) 60–80° (b) 90–100°

(c) 110–120° (d) 150–170°

Q110. The minimum foot-to-leg angle is:

(a) 45° (b) 60°

(c) 90° (d) 120°

Q111. Minimum fore-and-aft seat adjustment is:

(a) 50 mm (b) 100 mm

(c) 150 mm (d) 300 mm

Q112. Minimum vertical seat adjustment is:

(a) 25 mm (b) 100 mm

(c) 200 mm (d) 400 mm

Q113. Armrests must be:

(a) Fixed (b) Vertically adjustable for varying operator sizes

(c) Removed (d) Padded only

Q114. Operator's seat DIMENSIONS are given in:

(a) SAE J899 (b) SAE J209

(c) ASAE S335.3 (d) ISO 500

Q115. Zones of comfort and reach for control location are defined by:

(a) SAE J898 (b) SAE J185

(c) IS 4468 (d) ASAE S383.1

Q116. Dimensions for seat location relative to platform and controls:

(a) SAE J1163 (b) SAE J154a

(c) ISO 730 (d) IS 9020

Q117. Steps, handrails and handholds are covered by:

(a) SAE J185 (b) SAE J899

(c) ASAE EP443 (d) ISO 2631

Q118. The minimum operating 'space envelope' around the operator in a cab is guided by:

(a) SAE J154a (b) SAE J209

(c) IS 6284 (d) ASAE S304.5

Q119. Undamped natural frequency of a seat suspension should be:

(a) <1.5 Hz (b) 3–5 Hz

(c) 8–10 Hz (d) 20 Hz

Q120. Seat natural frequency should be what fraction of tyre natural frequency?

(a) 0.1 (b) 0.4–0.5

(c) 1.0 (d) 2.0

Q121. Maximum seat suspension travel is about:

(a) 25 mm (b) 76 mm

(c) 150 mm (d) 300 mm

Q122. Target seat isolation performance (transmissibility) is:

(a) 0.05 (b) 0.4–0.5

(c) 1.0 (d) 2.0

Q123. The penalty of a very soft seat suspension is:

(a) Poor isolation (b) Larger operator displacement relative to the controls

(c) Higher cost only (d) Noise

Q124. Static deflection due to operator weight is compensated by:

(a) Torsion bar or pneumatic cylinder (b) Extra padding

(c) Steel plate (d) Nothing

Q125. For best ride comfort, tyres should be:

(a) Smallest diameter (b) Largest diameter

(c) Solid (d) Narrow

Q126. Frequently used hand controls should be located:

(a) To the operator's LEFT (b) To the operator's RIGHT, leaving the left hand for steering

(c) Behind the seat (d) On the floor

Q127. Which controls are assigned OUTSIDE the primary zone (left hand/foot)?

(a) Hand throttle (b) Shift levers

(c) PTO clutch, parking brake, differential lock (d) Rock-shaft lever

Q128. Uniform location and direction of motion of operator controls is specified in:

(a) ASAE S335.3 (b) SAE J899

(c) ISO 2631 (d) IS 9129

Q129. Per that standard, a foot clutch pedal shall be operated by the:

(a) Right foot, upward
(b) Left foot, forward and/or downward to disengage
(c) Either foot
(d) Hand

Q130. Universal symbols for operator controls are published in:

(a) ASAE S304.5 / SAE J389b
(b) SAE J185
(c) IS 4931
(d) ISO 11783

Q131. Colour coding of hand controls is recommended in:

(a) ASAE EP443
(b) SAE J209
(c) ISO 730
(d) IS 6316

Q132. Controls used blind should be distinguishable by:

(a) Colour only
(b) Shape coding (and position)
(c) Weight
(d) Sound

Q133. Instrument face design and location is covered by:

(a) SAE J209
(b) SAE J898
(c) ASAE S310.3
(d) ISO 500

Q134. Highest-priority instruments should lie within the eye-movement cone of:

(a) 10°
(b) 30°
(c) 60°
(d) 90°

Q135. That priority zone is how wide at 700 mm from the operator's eye?

(a) 150 mm
(b) 280 mm
(c) 380 mm
(d) 600 mm

Q136. Per Purcell's panel rules, engine gauges are placed:

(a) Right of panel centre
(b) Left of panel centre
(c) Behind
(d) On the fender

Q137. Transmission gauges are placed:

(a) Left of centre
(b) Right of centre
(c) Below the seat
(d) Outside

Q138. Primary tractor visibility requires especially:

(a) Near-ground vision to front and rear
(b) Sky vision
(c) Under-engine view
(d) Rear-wheel view only

Q139. Because tractors are marketed worldwide, designers must use:

(a) Only US data
(b) Ethnic (population-specific) anthropometric data
(c) Averages only
(d) No data

Q140. Hansson et al. require that the operator should be able to:

(a) Stand throughout
(b) Change working position easily, with the area free of sharp edges/obstructions
(c) Work without instruments
(d) Reach all controls with feet

Answers

Q106 — (b) 5° with front raised — *Liljedahl pp. 226–227 — memorise 5°, 95°/90°, 110–120°, ≥ 90°, 150 mm, 100 mm.*

Q107 — (c) 95° (minimum 90°)

Q108 — (b) Lumbar back

Q109 — (c) 110–120°
Q110 — (c) 90°
Q111 — (c) 150 mm
Q112 — (b) 100 mm
Q113 — (b) Vertically adjustable for varying operator sizes
Q114 — (a) SAE J899 — *p. 227 — SAE family: J899 dimensions, J1163 location, J898 comfort/reach zones, J185 access, J154a space envelope.*
Q115 — (a) SAE J898
Q116 — (a) SAE J1163
Q117 — (a) SAE J185
Q118 — (a) SAE J154a
Q119 — (a) <1.5 Hz
Q120 — (b) 0.4–0.5
Q121 — (b) 76 mm
Q122 — (b) 0.4–0.5
Q123 — (b) Larger operator displacement relative to the controls — *p. 224 — the softness/displacement compromise.*
Q124 — (a) Torsion bar or pneumatic cylinder
Q125 — (b) Largest diameter — *p. 224 — the largest tyres filter terrain best.*
Q126 — (b) To the operator's RIGHT, leaving the left hand for steering — *p. 229 — right-hand primary zone; the left hand steers.*
Q127 — (c) PTO clutch, parking brake, differential lock
Q128 — (a) ASAE S335.3
Q129 — (b) Left foot, forward and/or downward to disengage — *p. 228 — clutch-pedal convention in ASAE S335.3.*
Q130 — (a) ASAE S304.5 / SAE J389b
Q131 — (a) ASAE EP443
Q132 — (b) Shape coding (and position)
Q133 — (a) SAE J209
Q134 — (b) 30° — *p. 229 — 30° easy-eye-movement cone.*
Q135 — (c) 380 mm — *2 × 700 × tan 15° ≈ 375 ≈ 380 mm.*
Q136 — (b) Left of panel centre
Q137 — (b) Right of centre
Q138 — (a) Near-ground vision to front and rear
Q139 — (b) Ethnic (population-specific) anthropometric data
Q140 — (b) Change working position easily, with the area free of sharp edges/obstructions